3a. $\Delta y = \dfrac{\lambda D}{d}$

$= \dfrac{vD}{fd}$

Frequency of sound, $f = \dfrac{vD}{d\,\Delta y}$

$= \dfrac{340 \times 12}{2.5 \times 0.60}$

= 2720 Hz

bi. $2A$

ii. $\dfrac{A}{\sqrt{2}} = 0.707\,A$

iii. $\left(A + \dfrac{A}{\sqrt{2}}\right) = 1.7071A$

$\approx 1.71A$

iv. Old intensity: $I = k(2A)^2$, k = constant

$= 4kA^2$

New intensity: $I' = k(1.7071A)^2$

$= 2.91kA^2$

$= 0.729\,I$